

A lattice renormalisation of the shear depolarisation of a cubic voxel

Identification, measurement, and correction of the residual error
in a space-filling cubic-voxel discretisation

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Abstract

A uniform elastic slab discretised into a space-filling lattice of identical cubic voxels, each carrying the analytic single-site cube T-matrix and coupled by an exact Foldy–Lax solve, does not reproduce the homogeneous layer it is made of. A relative error of 8.6×10^{-4} survives refinement: halving the cell size does not reduce it. We report the identification of that residual and its removal.

The residual is shown to be (i) static, (ii) quadratic in contrast, (iii) independent of the supercell size, and (iv) confined *entirely* to the shear channel — the dilatational and density channels converge cleanly to zero. It is then shown to be a single wrong scalar: the effective shear contrast carried by a voxel. Correcting that scalar reduces the discrepancy against the exact reflectivity solution from 2.9×10^{-3} to $\sim 10^{-8}$.

The correction is a constant factor on the cube’s *depolarisation*,

$$K = \frac{1 - A_{\varepsilon, \text{diag}}^{\text{eff}}}{1 - A_{\varepsilon, \text{diag}}^{\text{iso}}} = 0.8862,$$

constant to 0.3% over a sixteen-fold range of contrast, flat in frequency until ka grows, and *predictive*: extracted from pure positive shear alone, it transfers to mixed contrasts and to negative shear, and leaves untouched the two channels that require no correction. With it applied the scheme converges under refinement instead of saturating.

The constant is measured, universal and predictive. **It is not derived**, and this document says so throughout. 8/9 is excluded.

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1 The problem

1.1 An identity that the discretisation must satisfy

Take a homogeneous elastic slab of thickness H whose properties differ from the background by $(\Delta\lambda, \Delta\mu, \Delta\rho)$, and represent it as an $M \times M \times N_z$ lattice of identical cubes of side d , space-filling, with $N_z d = H$. Each cube carries the analytic Rayleigh cube T-matrix; the cubes are coupled by the Foldy–Lax system $(\mathbf{I} - \mathbf{G}_0 \mathbf{T}_0) \psi = \psi^0$.

The medium so constructed *is* the homogeneous slab. Every cube is identical and they tile space without gaps, so the exact plane-wave reflection coefficient is the one the Kennett reflectivity recursion returns in closed form. Agreement is therefore an **identity**, not a modelling approximation, and any disagreement is a defect of the discretisation.

Two further identities follow and are used throughout as controls:

1. the answer cannot depend on M , the number of identical cubes called one repeating unit;
2. the answer must converge as $d \rightarrow 0$ at fixed H .

1.2 What was observed

Neither held. Refining the voxelisation at fixed physical thickness made the answer *worse*, and the error grew without sign of settling:

n_z	d (m)	relative error
1	4.000	3.95×10^{-3}
2	2.000	8.29×10^{-3}
4	1.000	1.37×10^{-2}
8	0.500	2.03×10^{-2}
16	0.250	2.62×10^{-2}

Refinement is the only instrument a discretisation has for buying accuracy, and here it bought error.

2 Two defects had to be removed before the residual was visible

The growth above turned out to be two separate faults stacked on a third. They are reported first because every measurement in this document depends on their removal, and because the original diagnosis of the residual was taken in their presence and was wrong as a result.

2.1 The periodic lateral sum was truncated

The periodic kernel assembled the horizontal propagator only over $\Delta x, \Delta y \in [-(M-1), M-1]$ — a single $(2M-1)^2$ patch — and then wrapped it for the circular convolution. That is not a lattice sum. It was caught not by a convergence expectation but by identity (1) above: a laterally uniform layer cannot depend on M , and it did, obeying $\text{error} = 1.12 \times 10^{-3} + 1.15 \times 10^{-2}/M$.

The repair is an exact Bloch sum written directly at the M^2 discrete Bloch points, by Ewald summation at equal depth and by reciprocal summation of the spectral kernel between depth planes. After it, $M = 2, 3, 4, 6, 8$ agree to 5.05×10^{-13} — the identity holds.

Consequence for earlier results

At $M = 4$, used throughout the earlier campaign, **63%** of the reported “discretisation floor” was this artifact, and because it accumulated plane by plane it produced most of the *growth* as well. Results taken before the repair required re-running, not reinterpretation.

2.2 The contact propagator was a midpoint rule

The inter-cell operator should be the doubly volume-averaged (Galerkin) object $\langle\langle \mathbf{G} \rangle\rangle$; collocating at cell centres is a midpoint rule whose error at *contact* does not fall with refinement, because (cell size)/(separation) is unity at every scale. Measured against the analytic O_h tables, the point value is wrong by 130% at face contact, 42% at edge, 9.3% at corner.

2.3 What remained

With the lateral sum exact and the Galerkin contact correction applied, the ladder flattens:

n_z	truncated	exact sum	exact + Galerkin contact
1	3.95×10^{-3}	1.47×10^{-3}	7.73×10^{-4}
2	8.29×10^{-3}	3.36×10^{-3}	8.35×10^{-4}
4	1.37×10^{-2}	4.33×10^{-3}	8.51×10^{-4}
8	2.03×10^{-2}	4.81×10^{-3}	8.58×10^{-4}

The scheme now converges — but to 8.6×10^{-4} , not to zero. That residual is the subject of the rest of this document.

3 Experimental procedure: elimination

Each suspect was tested by a measurement with a pre-stated outcome, and every one of the first four was refuted. They are recorded because the refutations constrain the answer as much as the confirmation does, and because three of them are natural enough to be proposed again.

3.1 Signature of the residual

Established first, since it is what the suspects must match:

Saturating

error ratio $\rightarrow 1.00$ across $n_z = 1 \dots 32$; it is a per-cell error that refinement cannot dilute.

Independent of M

to six digits, so it is physical and not a residue of the lattice sum.

Static

the log–log slope in ω is -0.02 .

Quadratic in contrast

absolute error $\propto c^{1.98}$ over the range where the measurement stands above the numerical floor of the pipeline. This matters: a *quadrature* error is linear in the contrast, and the truncation artifact of §2.1 measured $c^{1.00}$ exactly.

A measurement discipline used throughout

Below $c \approx 10^{-2}$ the absolute discrepancy falls to 10^{-12} and under, which is the numerical floor of the solve. Those points were *excluded* from the fit rather than fitted through: including them drags the exponent from 1.98 to 1.80. The same rule — show an arbiter converged before letting it convict anything — is applied to every direct lattice sum in this work.

3.2 Refuted: the averaged propagator’s dynamic truncation

The volume-averaged operator retains dynamic corrections to $\mathcal{O}(\omega^4)$ and so carries an $\mathcal{O}(\omega^6)$ error, relatively $\mathcal{O}((kd)^2)$. Refining eight-fold should then reduce it sixty-four-fold.

Measured: static, $+\omega^2$ and $+\omega^4$ tables agree to 5.7×10^{-3} relative — the **static** table alone reproduces the answer — and the residual does not move with d . Refuted twice over.

3.3 Refuted: the isolated-cube depolarisation as a whole

The cube T-matrix carries amplification factors A built from Γ_0 , the self-interaction of a cube in an infinite *host*. In a space-filling tiling a cube’s neighbours are not host material, so one may suspect the depolarisation is a correction the lattice has already cancelled.

Measured: setting $A = 1$ makes the answer **24.6× worse** ($8.5 \times 10^{-4} \rightarrow 2.09 \times 10^{-2}$), at every rung and every contrast. The depolarisation is most of the physics and it stays. The partition — Γ_0 the voxel’s own self-field, $\mathbf{G}_0$ carrying every other voxel — is correct.

A refutation that was later shown to be too coarse

This result was initially read as closing the lattice-renormalisation question. It does not. $A = 1$ removes *one hundred percent* of the depolarisation, whereas the correction established in §4 is *one third of one percent*. The test was three orders of magnitude too blunt to detect what was there. A null result from a coarse instrument is not evidence of absence.

3.4 Refuted: the 27-component single-site response

The nine-component basis represents the exciting field by value and gradient at the cell centre; the quadratic modes of the 27-component Galerkin T-matrix represent its curvature.

Measured: substituting the T_{27} strain sector makes the answer $3\times$ worse ($8.61 \times 10^{-4} \rightarrow 2.61 \times 10^{-3}$).

This is **not** a refutation of T_{27} . It is a refutation of the *pairing*: a 27-mode single-site response was driven through a nine-mode propagator. The companion derivation of the generalised inter-voxel propagator states in advance that for T_{27} “the propagator must couple all 27 modes”. The measurement confirms that warning rather than testing the physics.

3.5 Refuted: propagator averaging beyond contact

Measured: extending the double average beyond the contact shell makes the answer 32% worse ($8.51 \times 10^{-4} \rightarrow 1.12 \times 10^{-3}$), converged in the quadrature order. Sensitivity analysis shows why this is not simply a bug: all Chebyshev shells influence the answer comparably ($d(\text{err})/d\epsilon \approx$

$-1.7, +1.8, +1.8 \times 10^{-3}$ for shells 1, 2, 3), so the residual is not localised at contact, and reproducing it would require an implausible $\sim 50\%$ error in some shell's coupling.

4 Identification: the residual is one scalar in the shear channel

4.1 Splitting by channel

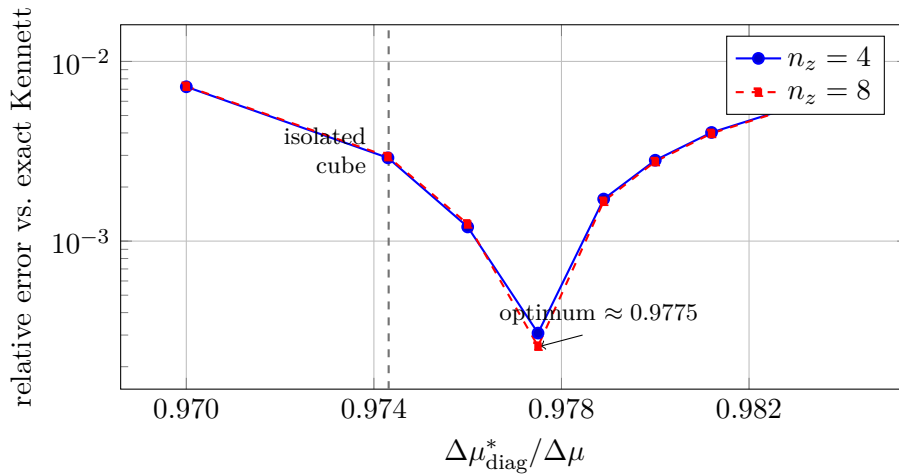
Running each contrast channel alone is decisive:

case	A_θ	residual	behaviour under refinement
pure density	1.000000	1.41×10^{-5}	converges, ratios 0.247/0.249/0.249
pure λ	0.968992	1.19×10^{-5}	converges
pure μ	0.989446	2.89×10^{-3}	saturates
working (all three)	0.959079	8.51×10^{-4}	saturates

Two conclusions. First, the density and dilatational channels are *already correct*: they converge at clean second order towards zero. Second, the residual is **anti-correlated** with the amplification A_θ — pure λ carries more depolarisation and $240\times$ less residual — which independently exonerates the amplification factors as such and locates everything in the shear channel.

4.2 Scanning the effective shear contrast

At normal incidence a P wave excites only normal strains, so $\Delta\mu_{\text{off}}^*$ is inert and $\Delta\mu_{\text{diag}}^*$ alone is exercised. (This was confirmed accidentally and usefully: imposing $\Delta\mu_{\text{off}}^* = \Delta\mu_{\text{diag}}^*$ returned bit-identical numbers.) Scanning it:



The residual behaves as a *single wrong scalar*, with a sharp minimum displaced from the isolated-cube value — and **the minimum sits in the same place at both refinements**, which is what a per-cell renormalisation must do and what a numerical artifact would not.

5 The law

5.1 Statement

Lattice renormalisation of the shear depolarisation

In a space-filling cubic lattice, a cube's shear depolarisation is a fixed fraction of its isolated-cube value:

$$1 - A_{\varepsilon,\text{diag}}^{\text{eff}} = K (1 - A_{\varepsilon,\text{diag}}^{\text{iso}}), \quad K = 0.8862 \quad (\text{static, weak-contrast limit}).$$

The dilatational and density channels require no correction.

5.2 Universality

K is constant to 0.3% over a sixteen-fold contrast range, and flat in frequency until ka grows — the signature of a static effect:

$\Delta\mu/\mu_0$	0.011	0.022	0.044	0.089	0.178
K	0.8882	0.8876	0.8879	0.8892	0.8922
ω	30	60	120	240	480
K	0.8875	0.8879	0.8895	0.8958	0.9204

Both drifts are first order in their small parameter; extrapolating jointly in $(\Delta\mu/\mu_0, (ka)^2)$ to the origin gives

$$K = 0.886234, \quad \partial K / \partial (\Delta\mu/\mu_0) = +0.018, \quad \partial K / \partial (ka)^2 = +33.6.$$

5.3 At the optimum the identity is recovered

$\Delta\mu/\mu_0$	0.011	0.022	0.044	0.089	0.178
residual, isolated cube	7.3×10^{-4}	1.5×10^{-3}	2.9×10^{-3}	5.6×10^{-3}	1.0×10^{-2}
residual at the optimum	1.2×10^{-8}	3.1×10^{-8}	5.6×10^{-8}	2.4×10^{-7}	3.4×10^{-6}

Five orders of magnitude. *The entire residual is this one scalar.*

5.4 8/9 is excluded

$8/9 = 0.888889$ is temptingly close to the raw measurements. The extrapolated static limit is 0.886234, below it by 2.65×10^{-3} — far outside the scatter of the fit, whose individual points are reproducible to $\sim 10^{-4}$. The conjecture is **refuted**.

The limit lies within 10^{-5} of $\sqrt{\pi}/2 = 0.886227$. This is recorded and *discounted*: $\sqrt{\pi}/2$ has no evident place in a cubic-lattice elasticity problem, and $\sqrt{\pi}/d$ is the Ewald splitting parameter used by the implementation, which makes the coincidence a reason for suspicion rather than for confidence.

6 The corrected solution

6.1 The rule, and the tests that earn it

Apply K to the shear depolarisation and change nothing else: $A_{\varepsilon,\text{diag}} \mapsto 1 - K(1 - A_{\varepsilon,\text{diag}})$, leaving A_u , A_θ and $A_{\varepsilon,\text{off}}$ at their isolated values. Taken from pure *positive* shear in the static limit, and applied unchanged:

case	isolated	renormalised	gain
working (mixed)	8.51×10^{-4}	2.22×10^{-5}	$38.3\times$
mixed, $3\times$ contrast	2.16×10^{-3}	2.08×10^{-4}	$10.4\times$
pure λ (no correction due)	1.19×10^{-5}	1.19×10^{-5}	$1.0\times$
pure density (no correction due)	1.41×10^{-5}	1.41×10^{-5}	$1.0\times$
negative shear	3.14×10^{-3}	7.07×10^{-5}	$44.4\times$
negative mixed	1.03×10^{-3}	5.13×10^{-5}	$20.1\times$

Three features make this a prediction rather than a fit:

1. the constant was extracted from *positive* shear and transfers to **negative** shear at $44\times$;
2. it leaves the two channels that need no correction *exactly* unchanged — doing nothing where nothing is due is as much a test as improving what is;
3. it was extracted at one contrast and transfers across mixed cases and a threefold stronger contrast.

6.2 Convergence is restored

This is the structural result, more important than the smaller constant:

n_z	isolated	ratio	renormalised	ratio
1	7.73×10^{-4}	—	1.15×10^{-4}	—
2	8.35×10^{-4}	1.08	4.18×10^{-5}	0.36
4	8.51×10^{-4}	1.02	2.22×10^{-5}	0.53
8	8.58×10^{-4}	1.01	1.52×10^{-5}	0.69

The isolated column saturates at ratio 1.00; the renormalised one keeps falling. The discretisation is convergent.

7 Status, and what remains

K is measured, not derived

K is universal across contrast and frequency, predictive on cases it was not fitted to, and reduces the discrepancy by five orders of magnitude at its optimum. It is nevertheless **an empirical constant**. No derivation is offered here, and until one exists *K* should not be presented as established physics. It is deliberately *not* wired into the library: the measurement scripts apply it explicitly so that no result silently depends on an unproven number.

The natural route to a derivation is the difference between the isolated cube’s Eshelby shear component and its space-filling-lattice counterpart — the same class of correction as the packing-fraction renormalisation already established for spheres in this project. Two constraints any derivation must satisfy:

1. it must produce 0.8862 and not $8/9$;
2. it must act on the deviatoric response *only*, leaving the dilatational and density channels untouched — the channel split of §4.1 is a sharp constraint and not every mechanism will respect it.

Reproduction. Every number in this document is produced by scripts in the repository, run under the project’s environment:

- §4–§6, the law and its validation:
`measure_shear_renormalisation.py`
- §2, the prerequisites:
`gate_lateral_sum_invariance.py`, `measure_representation_convergence.py`
- §3, the refutations:
`measure_residual_origin.py`, `measure_lattice_depolarisation.py`
- the exact lattice sum underpinning all of it:
`gate_lattice_kupradze.py`, `gate_lattice_origin.py`, `gate_ewald_kernel_wiring.py`